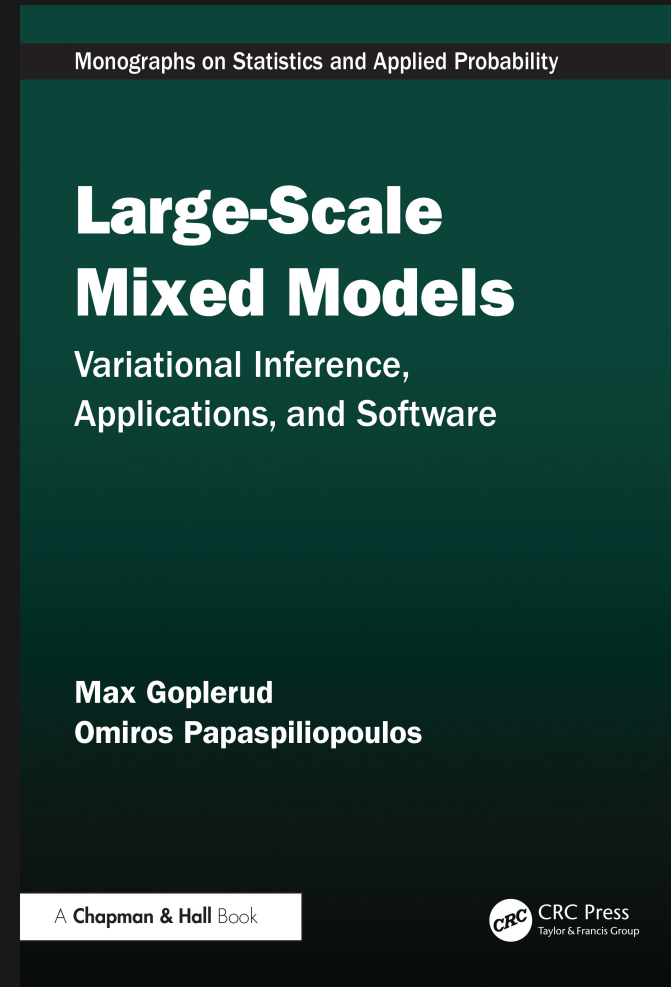


Part I

Intro and modeling framework

Acknowledgements

- Material relates to a book project
- **R Software**: vglmer
 - GLM-type model specification
 - Scales to thousands of parameters and millions of observations with considerable theoretical guarantees for computational performance and uncertainty quantification



- Course is structured around parts of the book
 - Introduce the broader modeling framework
 - Stand-alone mini-course on variational inference (VI)
 - VI for this class of models (theory and methods)
- Feedback very helpful and will be acknowledged
- Evaluation

GBMMs: big picture

Generalized bilinear mixed models (GBMMs)

Expressed in what we call “general design” notation:

$$y_i \mid \eta_i \sim f(y_i \mid \eta_i); \quad \eta_i = \mathbf{x}_i^T \boldsymbol{\beta} + \sum_{k=1}^K \mathbf{z}_{i,k}^T \boldsymbol{\alpha}_k + \sum_{b=1}^B \mathbf{u}_b^T \mathbf{Z}_{i,b} \mathbf{v}_b$$
$$i = 1, \dots, n$$

- $\mathbf{z}_{i,k}$: sparse design vectors
- $\mathbf{Z}_{i,k}$: sparse design matrices
- $\boldsymbol{\beta}$: fixed effects
- $\boldsymbol{\alpha}_k$: random coefficients
- $\mathbf{u}_b, \mathbf{v}_b$: random factors

- fixed effects
 - different treatments by different schools
- random coefficients:
 - random intercepts and slopes
 - associated with the random linear term in this additive model
- random factors:
 - also called latent factors
 - associated with the random bilinear term in this additive model
- we call collectively all random terms in the model random effects but also use the term latent variables
- general design notation is compact but unintuitive, we will consider more interpretable alternatives

Special cases

- GLMMs with random intercepts, random slopes and multi-way interactions
- GAMs
 - Including splines etc
- Item Response Theory/Factor Analysis
 - Including the main models used in psychology, political science, demography
- Probabilistic matrix factorization
- Models for dyadic and network data

A perspective

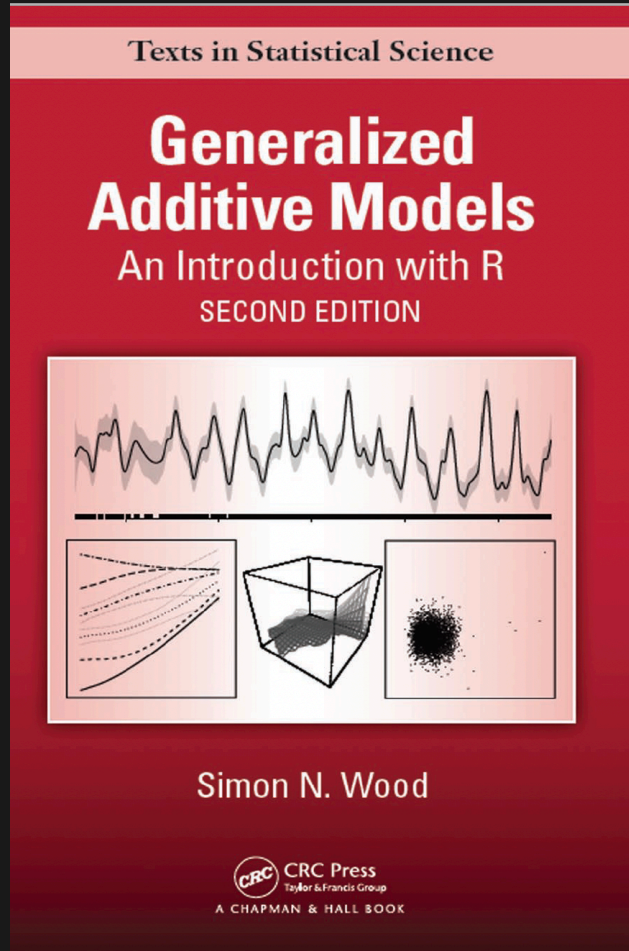
- this is a structured, flexible and interpretable framework for working with **high-dimensional categorical data**

GLMs, linear predictors,
varying coefficients and
interactions

Setting

- We start with the basic model consisting of **fixed effects** only
- Therefore:
 - We have a number n of observations, indexed by i , an output variable y_i and a number of covariates, some of which are **categorical**

- Good introductory reference (e.g., Chapter 1&2)



Motivating application: voter turnout

- This is an example that we have used in various works, and it is representative of a fundamental data structure and inference objective in political science
 - It is a **post-electoral survey** and predictive modeling is combined with **post-stratification**
 - These are described in detail here:

Deep Interactions with MRP: Election Turnout and Voting Patterns Among Small Electoral Subgroups

Yair Ghitza Columbia University
Andrew Gelman Columbia University

- The output $y_i \in \{0, 1\}$ indicates whether the i th respondent turned up to vote in previous election for about 80K respondents
- There are three continuous covariates - they are all defined at a state level (hence are the same for all people in a given state)
 - state-level income (**z_incstt**)
 - state-level % of republican candidate in previous election
 - state-level % of turnout in previous election (**z_trnprv**)
- Such covariates are known as “contextual”, especially when they correspond to a geographical unit

- There are three **ordinal** covariates:
 - personal income level
 - education level
 - age level (**age**)
 - e.g., in this dataset it has four *levels* (age groups): “18-29”, “30-44”, “45-64”, and “65+”
- Note that such variables are often treated as categorical, but also as numerical (usually once standardized) and in models in this context both of these practices are applied (e.g., income in both ways)

- There are several **categorical** covariates:
 - ethnicity
 - sex
 - region
 - state (**stt**)

- The goal is to build a predictive model that predicts whether a **voter type** will turn up to vote in the forthcoming election
 - Operationally, we train a regression model with y_i as output that uses the available covariates and features created from those
 - To see why the model predicts behavior of a “voter type” note that by construction the predictor will be equal for all people in a given subgroup of the population (residents in a given state, of given income level, age group etc)

- The actual values that a categorical/ordinal variable takes are meaningless
 - This is usually reflected in how the data are stored (although, not a good practice) where one binary might be recorded as 0/1 and another 1/2 (it happens in Application 1)
- The only thing that matters is that they correspond to different categories - when ordinal their order matters
 - The different values a categorical variable takes are called **levels**
 - E.g., in Application 1 **age** has 4 levels and **stt** 51

- There are various reasons why many individual-level covariates are recorded in categorical/ordinal scales, as opposed to their exact numerical value
 - Privacy
 - Robustness
 - Nonparametric modeling (further explanation)
 - **Post-stratification** (this works together with a voter-type predictive model)
- It is fair to say that modern datasets abound with categorical variables
 - Think also of text/choices/activities

- Using such basic covariates, we can create more **features** by considering functions thereof
- Two foundational types of features in this context are:
 - **dummy variables**
 - **interactions**

GLMs

- A more informal description¹ of this class of models is:
 - The distribution of y_i belongs to a parametric family, and the mean parameter is transformed so that it takes values in the real line and it is modeled as a linear function of available features
 - The linear function is called the **linear predictor**
 - There might be additional parameters in the density, e.g., dispersion parameters

- A more restricted definition of a GLM is as follows:
 - The distribution of y_i belongs to the **natural dispersion exponential family** (e.g., Gaussian for numerical, Poisson for counts, binomial for counts with known upper bounds)
 - A specific/canonical function, called the **link function**, maps its mean to the linear predictor, e.g, identity for Gaussian, log for Poisson, logit/probit/complementary log-log for binomial

- There are is a fundamental advantage of this more restricted view on GLMs:
 - The resulting negative log-likelihood is **convex** (in the regression parameters)
- This has important consequences for
 - Developing statistical theory for MLE
 - (More relevant here) Efficient computational framework for estimating GLMs using convex optimization
 - What is used primarily (unless there are good reasons not to..) is a variation of Newton-Raphson, known as **Fisher scoring**, which replaces the Hessian by the Fisher information matrix

- In Application 1 we take:

$$y_i \sim \text{Ber}(\pi_i)$$
$$\log \frac{\pi_i}{1 - \pi_i} = \eta_i$$

- This is of course the **logistic link function**
- In this formulation, the larger η_i the higher the probability respondent i turns up to vote

The linear predictor

- The contribution of a numerical covariate, say x_i is straightforward: βx_i
- Incorporating categorical covariates requires a little more care
 - What we typically want to do is to set the linear predictor to a different value for each level of the categorical covariate

- Consider Application 1 and say $\tilde{\eta}_i$ is a linear predictor made from other covariates, and we wish to incorporate “age”
- What we would typically want to do (again, unless there are good reasons to do otherwise..) is:

$$\eta_i = \tilde{\eta}_i + \begin{cases} \beta_1 & \text{if } i \text{ in group 18-29} \\ \beta_2 & \text{if } i \text{ in group 30-44} \\ \beta_3 & \text{if } i \text{ in group 45-64} \\ \beta_4 & \text{if } i \text{ in group 65+} \end{cases}$$

- We refer to this setting as **varying intercepts**

- Notice that this is also an example where the covariate is actually ordinal
- This is the “nonparametric” dimension we gave earlier to the approach of recording numerical variables in categorical/ordinal scales
 - By allowing a different value for each level of the categorical covariate we effectively model the relation between the mean and the actual age (in this example) as **piecewise constant function** and can learn from data non-linear (in age) patterns

- When it is desired that the linear predictor has a different value for each level of a categorical covariate, which we have called **varying intercepts** (as in the previous example), this is commonly achieved using the so-called **one-hot encoding**; a small variation of the same idea is the so-called **dummy variables**
 - explain the difference shortly, although mainly cultural

- Say the original categorical covariate with G levels
- In the one-hot-encoding, it is represented by a vector $\mathbf{m}_i$, which has G elements, $m_{i,g} \in \{0, 1\}$ for all $g = 1, \dots, G$, and $m_{i,g} = 1$ iff observation i takes the level g in this covariate
- We refer to such created variables as the **one-hot-encoding features** associated to a categorical covariate
 - Effectively, a single categorical covariate with G levels is represented by G binary-valued numerical features

- If β denotes the vector of the G different intercepts to which we wish to set the linear predictor, then this is achieved by the following contribution:

$$\eta_i = \tilde{\eta}_i + \beta^T \mathbf{m}_i$$

- Notice that the **design vectors** $\mathbf{x}_i$ are **sparse**

Identifiability, pt 1

- Pretty much every linear predictor (again, unless there are very good reasons not to..) will include a global **intercept**, say β_0
- This immediately creates an issue when the model includes a categorical covariate with varying intercepts, since:

$$\eta_i = \beta_0 + \beta^T \mathbf{m}_i$$

is **overparametrized**

- In our running example:

$$\eta_i = \begin{cases} \beta_0 + \beta_1 & \text{if } i \text{ in group 18-29} \\ \beta_0 + \beta_2 & \text{if } i \text{ in group 30-44} \\ \beta_0 + \beta_3 & \text{if } i \text{ in group 45-64} \\ \beta_0 + \beta_4 & \text{if } i \text{ in group 65+} \end{cases}$$

- There is one too many parameters here, there is no way the data can inform all 5 parameters in this model

- This problem is more subtle than it first appears
 - It becomes increasingly more complicated when dealing with several categorical covariates (see below)
 - It is also relevant when we discuss later random effects
- We will discuss this further later, here we outline a couple of ideas that work with a single categorical covariate (and variations thereof are often adapted when dealing with many)

- One approach is to work with **dummy variables**
 - This usually refers to the situation where a categorical covariate with G levels is represented by $G - 1$ binary-valued numerical features, as opposed to G
 - One level is taken as a **pivot**
 - The global intercept then matches the linear predictor for that level and in this formulation the other varying intercepts measure the difference of the linear predictor for the corresponding level relative to pivot

- In our running example, say we take age group “18-29” as pivot
 - Effectively we set $\beta_1 = 0$ and the interpretation, say, of β_2 is now the difference in the propensity to go to vote for age group “30-44” relative to “18-29”
- This is the default option in **R** and sets the first (in the default order in the system) coefficient to 0

- Another is to add **sum-to-zero constraints**
 - In our running example this would be achieved by imposing the constraint

$$\sum_{g=1}^4 \beta_g = 0$$

- Both are examples of **identifiability constraints** and in particular **linear identifiability constraints**
- These eliminate from the objective function a hyperplane of constant value
 - Effectively, when optimizing the objective function they pick a single point in a hyperplane of optimal value
- Within GLMs linear constraints can be still dealt with convex optimization tools
- Depending on the constraint adopted the interpretation of the coefficients is relative to a specific reference group

Interactions

- Consider Application 1 but now we wish to incorporate two categorical factors: **age** (as before) and **state**
- Say $\mathbf{m}_{i,1}$ and $\mathbf{m}_{i,2}$ are the design vectors associated to **age** (4-dimensional) and **state** (51-dimensional)
- The linear predictor

$$\eta_i = \beta_0 + \beta_1^T \mathbf{m}_{i,1} + \beta_2^T \mathbf{m}_{i,2}$$

implies that the propensity to go to vote for an individual varies according to both their age group and their state

- However, the effect, say of state, adds to whatever is that of age
 - In other words, the effect of state applies in the same way to all age groups
 - If say $\beta_{2,g} < 0$ for some state g , then all age groups in that state would have a propensity to go and vote smaller than the reference group (this will depend on whatever identifiability constraints are imposed)

- Interactions assign a different parameter to each cell in the **contingency table** defined by the **crossing** of the two categorical factors
- Even without interactions the linear predictor registers a different value for each cell, but that value is determined by a row and a column effect
 - With interactions, each cell is modeled separately

- This, of course, comes at the cost of increasing number of parameters; for two categorical factors with G_1 and G_2 numbers of levels, an interaction introduces $G_1 \times G_2$ number of parameters (subject, though, to identifiability constraints)

- Operationally, interactions are constructed by creating a new categorical covariate with number of levels the product of the numbers of the constituent ones
 - We refer to this as the **outer product** of the two categorical covariates
 - They can be then represented using one-hot-encoding as any other categorical covariate

- Hence, from a mathematical/computational point of view interactions are effectively factors with large(r) number of levels
 - But! these new covariates have a specific type of linear dependence with the constituent variables, something that will turn out to be an issue

- We can also create an interaction between a numerical, say **z_trnprv**, and a categorical, say **age**, covariate
- Their interaction would give a different coefficient that multiplies **z_trnprv** for each level of **age**
 - In our example, their interaction could allow the higher the previous state turnout the less likely older voters will go to vote (negative slope) but more likely younger ones (positive slope)

- Operationally, this can be achieved by creating new numerical features as the product of the numerical covariate and the one-hot-encoding features that represent the categorical one
- We refer to this as **varying slopes** and to a model where there are both varying intercepts and slopes as a **varying coefficient** model
- There are identifiability issues also with varying slopes, as we will see next